

# Occlusion-Aware Fruit Instance Segmentation for Agricultural Robotics: A Comparative Study of YOLO11, YOLO26, and Mask2Former

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**Abstract.** Robust fruit perception in orchard environments remains challenging due to partial occlusions, illumination variability, and cluttered backgrounds. This work presents a comparative evaluation of CNN-based and transformer-based instance segmentation models for multi-fruit perception in agricultural robotics.

The study compares YOLO11-seg, YOLO26-seg, and Mask2Former under a unified training and evaluation protocol using a dataset of real fruit specimens captured in a controlled environment that emulates orchard-like foliage occlusion, spanning multiple visibility conditions. The analysis considers both segmentation accuracy and computational efficiency using metrics such as precision, recall, mAP, parameter count, FLOPs, and inference time.

Experimental results show that YOLO11m provides the best trade-off between accuracy and efficiency among the evaluated YOLO models, achieving 51.97% mask mAP<sub>50-95</sub>. In contrast, Mask2Former attains the highest overall segmentation performance with 61.01% mask mAP<sub>50-95</sub> and improved robustness under severe occlusion, at the cost of higher architectural complexity and computational requirements.

The results highlight the trade-off between real-time efficiency and segmentation robustness for robotic harvesting systems operating in occlusion-rich agricultural environments.

**Keywords:** Instance segmentation · YOLO · Mask2Former · Agricultural robotics · Fruit perception · Occlusion robustness

## 1 Introduction

Agricultural robotics has gained increasing attention for improving productivity and reducing labor dependency in precision agriculture applications [5, 23, 22]. In particular, autonomous fruit harvesting remains challenging due to the unstructured nature of orchard environments, where illumination changes, cluttered backgrounds, fruit overlap, and severe occlusions frequently affect visual perception systems.

Reliable fruit perception is essential for robotic harvesting, since detection and segmentation directly influence localization and manipulation performance. While bounding-box detection provides coarse localization, instance segmentation offers pixel-level object delineation, making it more suitable for robotic interaction in highly occluded environments.

Recent CNN-based architectures, particularly YOLO-based models, have demonstrated strong real-time performance for agricultural perception tasks [2, 16, 21, 11]. At the same time, transformer-based architectures such as Mask2Former have shown improved contextual reasoning and robustness in complex, cluttered segmentation scenarios outside agriculture – including fine-grained fish analysis, change detection, and stacked-object counting – suggesting that these benefits generalize beyond any single domain [1, 6, 20, 10].

Despite these advances, comparative analyses between modern YOLO-based and transformer-based instance segmentation models in orchard environments remain limited, especially under varying visibility conditions and occlusion levels.

To address this gap, this work presents a comparative evaluation of YOLO11-seg, YOLO26-seg, and Mask2Former for multi-fruit instance segmentation under orchard-like foliage occlusion. The evaluation considers both segmentation accuracy and computational efficiency under different visibility conditions.

## 2 Related Work

### 2.1 Fruit Perception Under Occlusion

Visual perception remains one of the main challenges for agricultural robotic harvesting systems operating in orchard environments. Factors such as foliage occlusion, overlapping fruits, illumination variability, and cluttered backgrounds significantly affect detection and segmentation performance [2, 16, 7].

Traditional computer vision methods based on handcrafted features have shown limited robustness under these conditions [8]. More recent deep learning approaches have considerably improved perception capabilities in agricultural environments; however, severe occlusion still causes incomplete detections, poor mask quality, and instance separation failures in dense scenes [11, 15].

Instance segmentation has therefore become increasingly relevant for robotic harvesting applications because it provides pixel-level object delineation, which is essential for localization and manipulation in partially visible scenarios.

## 2.2 YOLO-Based and Transformer-Based Segmentation Models

YOLO-based architectures are widely used in agricultural perception tasks due to their balance between accuracy and real-time performance [13, 21, 17]. Recent variants such as YOLO11 and YOLO26 introduce improvements in feature extraction, multi-scale fusion, and end-to-end prediction strategies aimed at enhancing efficiency and robustness [18, 9, 19].

Despite their efficiency, most YOLO-based agricultural studies focus primarily on bounding-box detection performance rather than segmentation robustness under occlusion conditions.

In parallel, transformer-based architectures have emerged as a powerful alternative for dense scene understanding and universal image segmentation. Mask2Former further advanced transformer-based segmentation through masked attention mechanisms for universal image segmentation tasks [4]. Recent studies outside agriculture have shown that transformer-based segmentation models achieve strong performance in cluttered scenes involving overlapping and visually similar objects [6, 1, 10]. By combining convolutional feature extraction with transformer attention mechanisms, Mask2Former improves contextual reasoning and mask refinement, making it particularly suitable for highly occluded agricultural environments.

## 2.3 Research Gap

Although both CNN-based and transformer-based architectures have shown promising results for agricultural perception, comparative studies between modern YOLO-based and transformer-based instance segmentation models remain limited, particularly under controlled occlusion conditions.

Furthermore, most previous works evaluate models using global metrics only, without explicitly analyzing segmentation stability across different levels of object visibility.

To address these limitations, this work presents a comparative evaluation of YOLO11-seg, YOLO26-seg, and Mask2Former under multiple visibility conditions, analyzing both segmentation quality and computational efficiency in realistic orchard environments.

# 3 Materials and Methods

This section describes the proposed perception pipeline, the instance segmentation model employed, the dataset construction, and the evaluation methodology used to assess performance under occlusion.

## 3.1 System Overview

The proposed pipeline is designed to support robust visual perception in agricultural environments characterized by clutter and partial visibility, as illustrated

in Fig. 1. Given an RGB image acquired from an onboard camera, the system performs instance segmentation of fruit objects, producing pixel-wise masks and associated confidence scores. The pipeline is intended for integration within a broader robotic framework, enabling downstream tasks such as grasp planning and manipulation.

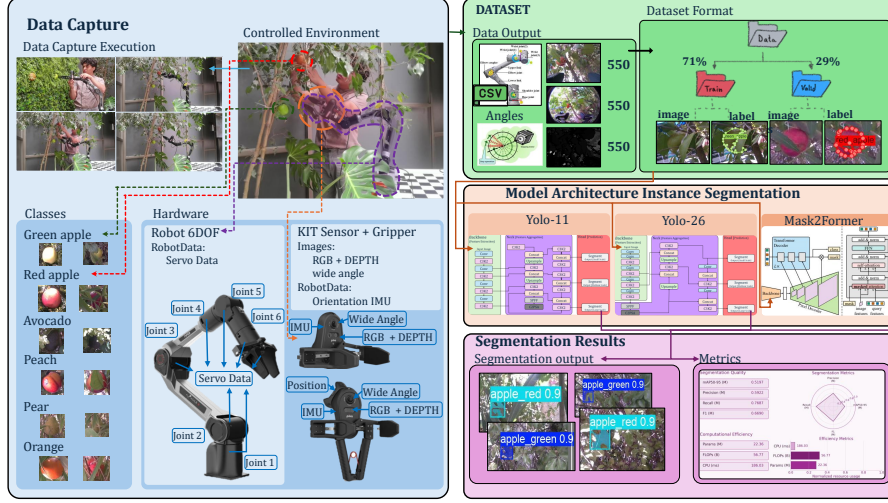


Fig. 1: Overview of the proposed pipeline: data acquisition setup using a robotic manipulator equipped with RGB and depth sensors, the resulting dataset format, and a high-level overview of the YOLO11, YOLO26, and Mask2Former architectures used for instance segmentation. Only the RGB images were used for the segmentation task reported in this work; depth and IMU data were collected for other purposes within the broader robotic framework and are not used here.

### 3.2 Data Acquisition Setup

Data were acquired using an RGB camera mounted on a robotic manipulator operating in a controlled environment. Real specimens of each fruit species were arranged among branches and foliage to emulate the partial occlusion conditions found in orchards, rather than being collected during harvesting in a working orchard; this controlled setting allowed the six fruit species considered in this work (which do not co-occur in a single real orchard) to be captured under comparable, repeatable conditions. The robotic platform allowed the camera to capture fruit images from multiple viewpoints while maintaining a consistent observation distance.

This setup enabled the collection of images with, predominantly, occlusion by foliage (branches and leaves); occlusion from neighboring fruit instances occurs

less frequently in this dataset, which averages 1.13–1.19 fruit instances per image (Table 1). The acquisition pipeline is illustrated in Fig. 1, where the robotic manipulator is used as a mobile sensing platform for collecting visual data used in training and evaluating the instance segmentation models.

### 3.3 Instance Segmentation Architectures

The proposed perception framework evaluates both convolutional and transformer-based instance segmentation architectures to analyze their robustness under occlusion-rich agricultural environments.

Two families of YOLO-based models were considered, namely YOLO11-seg and YOLO26-seg, representing efficient single-stage convolutional architectures designed for real-time instance segmentation. In addition, the transformer-based Mask2Former architecture with a ResNet-50 backbone was incorporated as a high-capacity segmentation baseline.

The YOLO-based models follow a unified single-stage paradigm in which object localization and mask prediction are jointly optimized within a single forward pass. As illustrated in Fig. 1, these architectures are composed of three main components: a convolutional backbone for hierarchical feature extraction, a multi-scale feature aggregation neck, and a segmentation head responsible for predicting both bounding boxes and instance masks.

The convolutional backbone extracts low-level spatial features and high-level semantic representations, while the neck integrates multi-scale information to improve robustness against scale variations and partial occlusions commonly observed in orchard environments. The segmentation head operates over the aggregated feature maps to generate pixel-wise masks associated with each detected fruit instance.

In contrast, Mask2Former adopts a hybrid CNN-transformer architecture designed for universal image segmentation. The model combines a convolutional backbone with transformer-based attention modules capable of modeling long-range spatial dependencies and contextual relationships between object regions. Instead of predicting masks directly from dense convolutional outputs, Mask2Former formulates instance segmentation as a mask classification problem, where learnable queries are iteratively refined through transformer attention mechanisms.

This transformer-based design enables stronger contextual reasoning under severe occlusion and overlapping object conditions, which are common in agricultural scenes containing foliage clutter and partially visible fruits. However, these benefits come at the cost of increased computational complexity and memory usage.

The inclusion of both YOLO-based and transformer-based architectures enables a comprehensive comparison between efficient real-time segmentation models and higher-capacity contextual reasoning approaches. This comparison is relevant for perception-driven robotic harvesting systems, where the balance between segmentation accuracy, robustness to occlusion, and computational efficiency directly affects robotic manipulation performance.

### 3.4 Training Configuration

All models were trained using the default settings of their respective frameworks, matched where the frameworks’ options allowed (optimizer, learning rate, batch size, input resolution); full equivalence was not possible since YOLO-seg and Mask2Former expose different training options, as detailed below. YOLO11-seg and YOLO26-seg were trained using the Ultralytics YOLO framework [12], whereas Mask2Former was implemented using the OpenMMLab MMDetection framework [3].

Training was performed using the AdamW optimizer with an initial learning rate of 0.001 and a batch size of 8, with an input image resolution of  $640 \times 640$  pixels. YOLO11-seg and YOLO26-seg were trained for 100 epochs, and the final-epoch checkpoint was used for evaluation in all cases to keep the reporting protocol identical across model sizes; Mask2Former was trained for 50 epochs, as further training did not improve validation performance, and the checkpoint with the best validation mask AP was used for evaluation. GPU latency (Table 4) was measured at batch size 1,  $640 \times 640$  input resolution, and default (FP32) precision for all models; the YOLO-based and Mask2Former latencies were obtained from each framework’s own benchmarking utility (Ultralytics and MMDetection, respectively), which do not necessarily time the same set of pre/post-processing steps, and should accordingly be treated as indicative rather than strictly apples-to-apples.

Data augmentation techniques including horizontal flipping, scaling, and color jittering were applied during training to improve generalization. For Mask2Former, multi-scale augmentation following the standard training protocol was additionally employed. This additional augmentation is part of Mask2Former’s standard MMDetection training recipe and was not replicated for the YOLO-based models; it therefore constitutes an asymmetry in the training protocol that should be considered when interpreting the absolute accuracy gap between Mask2Former and the YOLO-seg models, rather than attributing it solely to architectural differences.

The experiments were executed on a workstation equipped with an NVIDIA GeForce RTX 4060 Ti GPU (8 GB VRAM), allowing efficient training and evaluation of all architectures. Pretrained weights from the COCO dataset [14] were used for initialization to accelerate convergence during fine-tuning on the fruit dataset.

For the transformer-based model, Mask2Former with a ResNet-50 backbone was adopted as the reference configuration.

### 3.5 Dataset and Annotation Protocol

The dataset consists of RGB images of real fruit specimens acquired in a controlled environment, using an RGB camera mounted on a robotic manipulator (Section 3.2). Images contain multiple fruit classes including apples, peaches, avocados, pears, and oranges, arranged among foliage to produce varying levels of occlusion; occlusion from neighboring fruits is comparatively rare (Table 1).

Table 1: Dataset statistics and class distribution for training and validation splits.

| <b>Metric</b>                 | <b>Train</b> | <b>Validation</b> |
|-------------------------------|--------------|-------------------|
| Images                        | 1224         | 491               |
| Instances                     | 1382         | 584               |
| Avg. objects per image        | 1.13         | 1.19              |
| <b>Class distribution (%)</b> |              |                   |
| Apple (green)                 | 15.63        | 19.52             |
| Apple (red)                   | 16.06        | 16.95             |
| Peach                         | 13.39        | 16.44             |
| Orange                        | 15.27        | 10.27             |
| Pear                          | 20.55        | 15.92             |
| Avocado                       | 19.10        | 20.89             |

All images were manually annotated at the instance level, generating pixel-wise segmentation masks for each visible fruit. The annotations were performed using a polygon-based labeling protocol to ensure accurate object boundaries.

To evaluate model generalization, the dataset was divided into training and validation subsets only; no independent held-out test set was used (this is discussed as a limitation of the study in Section 5). Table 1 summarizes the number of images used for each dataset partition.

### 3.6 Evaluation Metrics

Model performance was evaluated using standard object detection and instance segmentation metrics, including Precision, Recall, F-score, Intersection over Union (IoU), and mean Average Precision (mAP). In addition, computational efficiency was analyzed using the number of parameters (Params), floating-point operations (GFLOPs), and GPU inference time.

The evaluation metrics used in this work are summarized in Table 2.

All metrics were computed separately for bounding boxes (B) and instance masks (M), enabling a detailed evaluation of both detection accuracy and segmentation quality under different visibility conditions. Precision, Recall, and F-score were computed using each framework’s own default confidence-score and IoU operating point (0.5 IoU, Ultralytics’ and MMDetection’s respective default confidence thresholds); as these defaults are not necessarily identical between the two frameworks, absolute Precision/Recall/F-score differences between Mask2Former and the YOLO-seg models should be interpreted with this in mind, whereas mAP, being threshold-integrated, is directly comparable.

Table 2: Evaluation and computational complexity metrics used in this work.

| Metric                       | Definition                                                                              |
|------------------------------|-----------------------------------------------------------------------------------------|
| Precision                    | $\frac{TP}{TP + FP}$                                                                    |
| Recall                       | $\frac{TP}{TP + FN}$                                                                    |
| F-score                      | $\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$ |
| IoU                          | $\frac{ A \cap B }{ A \cup B }$                                                         |
| Average Precision (AP)       | $AP = \int_0^1 P(R) dR$                                                                 |
| Mean Average Precision (mAP) | $mAP = \frac{1}{N} \sum_{i=1}^N AP_i$                                                   |
| $mAP_{50}$                   | AP computed at IoU = 0.5                                                                |
| $mAP_{50-95}$                | Mean AP over IoU thresholds from 0.5 to 0.95                                            |
| Params                       | Total number of learnable model parameters                                              |
| GFLOPs                       | Floating-point operations per forward pass                                              |
| GPU Time                     | Inference latency measured on GPU                                                       |

## 4 Experimental Results

### 4.1 Quantitative Results

The quantitative performance of the evaluated instance segmentation models on the validation set is summarized in Tables 3 and 4. The comparison includes YOLO11-seg, YOLO26-seg, and the transformer-based Mask2Former architecture with a ResNet-50 backbone.

Table 3 reports detection and instance segmentation metrics, whereas Table 4 summarizes computational efficiency indicators including GPU inference latency, model parameters, and FLOPs. Accuracy-related metrics are expressed in percentage (%), while inference latency is reported in milliseconds (ms).

Table 3 presents a quantitative comparison of detection and instance segmentation performance across all evaluated architectures. Among the YOLO-based models, YOLO11 variants achieve superior or more stable performance than their size-matched YOLO26 counterparts for the n, s, and m sizes, whereas YOLO26x (51.00% mask mAP<sub>50-95</sub>) outperforms both YOLO11l (47.93%) and YOLO11x (48.21%); the comparison between the two families is therefore size-dependent

Table 3: Quantitative performance comparison of YOLO-based and transformer-based instance segmentation models.

| Model           | Var. | mAP50-95<br>(B) | mAP50-95<br>(M) | Prec<br>(B)  | Rec<br>(B)   | F1<br>(B)    | Prec<br>(M)  | Rec<br>(M)   | F1<br>(M)    |
|-----------------|------|-----------------|-----------------|--------------|--------------|--------------|--------------|--------------|--------------|
| YOLO11          | n    | 54.98           | 50.26           | 58.50        | 82.16        | 68.34        | 58.30        | 81.93        | 68.12        |
|                 | s    | 55.43           | 50.79           | 59.15        | 77.59        | 67.12        | 59.06        | 77.45        | 67.01        |
|                 | m    | 57.41           | 51.97           | 59.35        | 77.05        | 67.05        | 59.22        | 76.87        | 66.90        |
|                 | l    | 53.18           | 47.93           | 57.28        | 76.37        | 65.46        | 57.28        | 76.37        | 65.46        |
|                 | x    | 53.44           | 48.21           | 57.57        | 75.06        | 65.16        | 57.57        | 75.06        | 65.16        |
| YOLO26          | n    | 51.43           | 47.03           | 56.41        | 74.59        | 64.24        | 55.96        | 73.97        | 63.71        |
|                 | s    | 55.19           | 50.40           | 56.82        | 73.44        | 64.07        | 56.73        | 73.30        | 63.96        |
|                 | m    | 54.80           | 50.15           | 56.38        | 71.26        | 62.95        | 56.22        | 71.20        | 62.83        |
|                 | l    | 45.21           | 41.23           | 50.54        | 53.26        | 51.86        | 50.39        | 53.17        | 51.74        |
|                 | x    | 56.19           | 51.00           | 56.35        | 72.50        | 63.41        | 56.28        | 72.40        | 63.33        |
| Mask2<br>Former | R50  | <b>61.85</b>    | <b>61.01</b>    | <b>63.83</b> | <b>86.44</b> | <b>73.44</b> | <b>61.87</b> | <b>84.14</b> | <b>71.30</b> |

rather than uniform. In particular, YOLO11m attains the highest YOLO-based segmentation accuracy, reaching **51.97%** mask mAP50–95. YOLO26l is a clear outlier within its own family (41.23% mask mAP50–95 and 53.26% box recall, both well below every other YOLO26 size); its training curve shows this is not a failure to converge but the opposite problem, late-stage overfitting: validation mask mAP50–95 already reached 50.0% by epoch 15 and then declined steadily to 41.23% by epoch 100, the final-epoch checkpoint used for evaluation. The other YOLO26 sizes did not exhibit this degradation over the same 100 epochs, so the anomaly reflects this specific run’s optimization trajectory rather than a genuine architectural effect of model size.

When compared with the transformer-based baseline, Mask2Former achieves the highest overall segmentation performance, reaching **61.01%** mask mAP50–95 together with the highest recall values. These results suggest that transformer-based contextual reasoning improves instance separation and robustness under severe occlusion and cluttered orchard conditions.

Table 4 complements this analysis by reporting computational complexity and GPU inference latency. Although Mask2Former achieves the best overall segmentation performance, it requires substantially higher computational resources, with 44.00M parameters and 103.00B FLOPs. In contrast, YOLO11m retains 85.2% of Mask2Former’s mask mAP50–95 (51.97% vs. 61.01%) while using only 50.8% of its parameters (22.36M vs. 44.00M) and 55.1% of its FLOPs (56.77B vs. 103.00B), at a lower measured GPU latency (14.53 ms vs. 16.70 ms). We use this accuracy-retained-per-unit-of-compute criterion to characterize YOLO11m as offering the best accuracy-efficiency trade-off among the evaluated models, rather than as a general, unquantified claim.

Overall, the results indicate that transformer-based architectures provide superior segmentation robustness and mask quality, whereas YOLO11 models offer

Table 4: Computational efficiency comparison of YOLO-based and transformer-based models.

| Model       | Var. | GPU Latency<br>(ms) | Params<br>(M) | FLOPs<br>(B) |
|-------------|------|---------------------|---------------|--------------|
| YOLO11      | n    | 12.39               | <b>2.84</b>   | <b>4.87</b>  |
|             | s    | 14.29               | 10.09         | 16.54        |
|             | m    | 14.53               | 22.36         | 56.77        |
|             | l    | 22.50               | 27.62         | 66.31        |
|             | x    | 34.53               | 62.06         | 148.53       |
| YOLO26      | n    | 12.53               | 3.06          | 5.09         |
|             | s    | <b>12.05</b>        | 11.44         | 18.55        |
|             | m    | 20.65               | 26.98         | 65.95        |
|             | l    | 23.42               | 31.38         | 75.15        |
|             | x    | 36.74               | 70.49         | 168.35       |
| Mask2Former | R50  | 16.70               | 44.00         | 103.00       |

a more favorable balance between segmentation accuracy, inference speed, and computational efficiency for embedded robotic deployment.

Figure 2 compares the training evolution of all evaluated YOLO11-seg and YOLO26-seg variants together with Mask2Former, in terms of bounding-box and mask performance metrics; YOLO11m-seg and YOLO26x-seg are highlighted with bold lines as the best-performing variant of each YOLO family. The results show that Mask2Former achieves the highest overall segmentation accuracy, reaching a mask mAP50–95 of approximately **61.0%**, surpassing both YOLO11m-seg and YOLO26x-seg. Additionally, Mask2Former exhibits smoother convergence behavior and maintains consistently higher recall values throughout training, suggesting improved robustness under cluttered and partially occluded orchard conditions.

## 4.2 Visibility and Occlusion Analysis

To analyze robustness under different occlusion conditions, additional experiments were performed by partitioning the validation set according to fruit visibility levels (25%, 50%, 75%, and 100%).

Visibility levels were assigned manually during annotation rather than computed from a geometric occlusion ratio: each validation image was inspected individually, and the proportion of the fruit’s surface left unoccluded by surrounding vegetation (branches and leaves) was visually estimated and assigned to the closest of four categories. An image was labeled 25% when foliage covered substantially more than half of the fruit, 50% when occlusion and visible surface were approximately balanced, 75% when the visible surface clearly exceeded the occluded portion, and 100% when the fruit was fully or almost fully visible. Table 5 reports the resulting number of validation images per level; an additional 27 validation images containing fruits spanning more than one visibility level

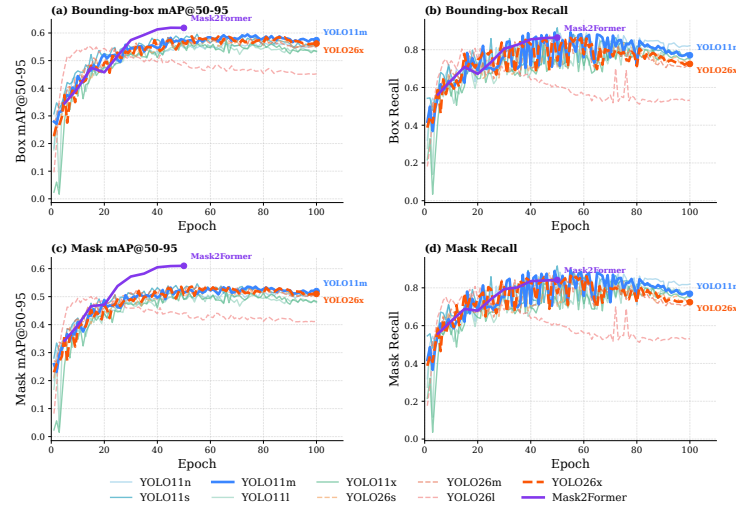


Fig. 2: Training evolution of detection and segmentation metrics for all evaluated models, including YOLO11, YOLO26, and Mask2Former. The figure reports (a) bounding-box mAP50–95, (b) bounding-box recall, (c) mask mAP50–95, and (d) mask recall across training epochs. Mask2Former achieves the highest overall segmentation accuracy and exhibits smoother convergence behavior, particularly in mask-related metrics. YOLO11m remains competitive while providing lower computational complexity and real-time inference capability.

Table 5: Number of validation images per fruit visibility level.

| Visibility level | 25% | 50% | 75% | 100% |
|------------------|-----|-----|-----|------|
| Images           | 142 | 94  | 133 | 95   |

within the same frame were kept aside as a separate “Multi” group and are not included in this single-level analysis.

Figure 3 presents the segmentation performance evolution for YOLO11m-seg, YOLO26x-seg, and Mask2Former across different visibility conditions. The results indicate that segmentation accuracy consistently improves as object visibility increases, which is expected due to the larger amount of observable visual information available to the models.

Among the evaluated approaches, Mask2Former demonstrates the highest robustness across all visibility levels, particularly under severe occlusions (25% and 50% visibility). The transformer-based architecture maintains more stable performance degradation compared to YOLO-based models, suggesting improved contextual reasoning and global feature aggregation.

YOLO11m-seg and YOLO26x-seg track each other closely under 25% and 50% visibility, with no consistent winner between them at these severe-occlusion levels. At 75% and especially 100% visibility, YOLO26x-seg reaches a higher

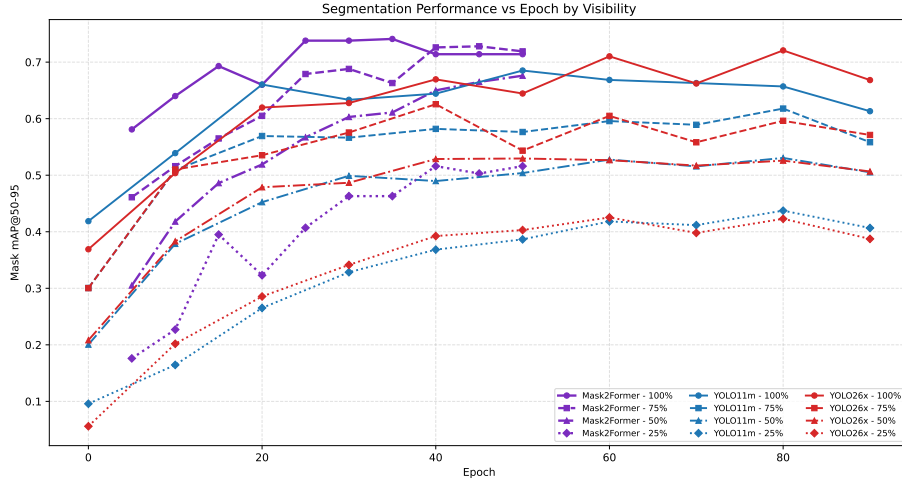


Fig. 3: Segmentation performance evolution under different visibility levels for YOLO11m-seg, YOLO26x-seg, and Mask2Former. Validation images were grouped according to object visibility percentages (25%, 50%, 75%, and 100%). Mask2Former demonstrates higher robustness under severe occlusions. YOLO11m-seg and YOLO26x-seg are closely matched at 25% and 50% visibility, while YOLO26x-seg reaches higher mask mAP50–95 than YOLO11m-seg at 75% and 100% visibility over most of training.

mask mAP50–95 than YOLO11m-seg over most of training, indicating that YOLO11m-seg’s advantage reported in Table 3 does not uniformly hold once the validation set is broken down by visibility level. This further supports the quantitative observations reported above, while also showing that YOLO26x-seg is the more robust YOLO-based choice specifically for images with higher fruit visibility.

### 4.3 Qualitative Comparison

Figure 4 presents a qualitative comparison between YOLO11m-seg, YOLO26x-seg, and Mask2Former using identical RGB inputs across multiple fruit categories and occlusion conditions. This analysis complements the quantitative evaluation by visually examining mask structure, contour consistency, and robustness under partial occlusion. The per-detection confidence scores overlaid on these predictions are produced by each framework’s own classification head and are not calibrated against one another; they should be read only as each model’s own relative confidence across its own detections, not compared numerically across architectures.

All evaluated models are capable of detecting the main fruit instances present in the scene. However, noticeable qualitative differences emerge in the shape, smoothness, and stability of the generated masks.

YOLO26x-seg generally produces valid detections with high confidence scores in relatively clear visibility conditions, but it more frequently exhibits mask fragmentation and irregular contours under severe occlusion. In several cases, the predicted masks present polygonal boundaries and reduced adherence to the actual fruit geometry, particularly in cluttered regions involving foliage and overlapping objects.

YOLO11m-seg provides more stable mask generation and improved robustness across challenging scenes compared to YOLO26x-seg. The predicted masks tend to remain more coherent under partial occlusion while preserving satisfactory instance separation and geometric consistency. These observations are consistent with the higher recall and segmentation performance reported in the quantitative analysis.

The transformer-based Mask2Former architecture produces the most visually refined segmentation results among all evaluated models. In particular, Mask2Former generates smoother and more geometrically consistent masks with improved adherence to fruit contours and fewer polygonal artifacts. This behavior is especially evident in highly occluded examples such as avocado, pear, and red apple, where the predicted masks remain continuous despite significant background clutter and partial visibility.

Additionally, Mask2Former consistently produced stable predictions across all evaluated fruit categories, even under partial occlusion and cluttered background conditions.

These qualitative observations are aligned with the quantitative results presented in Tables 3 and 4. While Mask2Former achieves the highest overall segmentation accuracy and superior robustness under occlusion, YOLO11m-seg offers a more favorable balance between segmentation quality and computational efficiency for real-time robotic applications.

From an operational perspective, coherent and stable instance masks are particularly important for downstream robotic tasks such as grasp planning, fruit localization, and collision-free manipulation. In this context, Mask2Former provides the best qualitative segmentation behavior, whereas YOLO11m-seg remains an attractive solution for embedded real-time perception systems requiring efficient onboard inference.

## 5 Discussion

The experimental results reveal clear differences between CNN-based and transformer-based instance segmentation architectures when applied to fruit perception in occlusion-rich agricultural environments. Although all evaluated models were capable of detecting and segmenting fruit instances, important differences were observed in segmentation stability, mask quality, and computational efficiency.

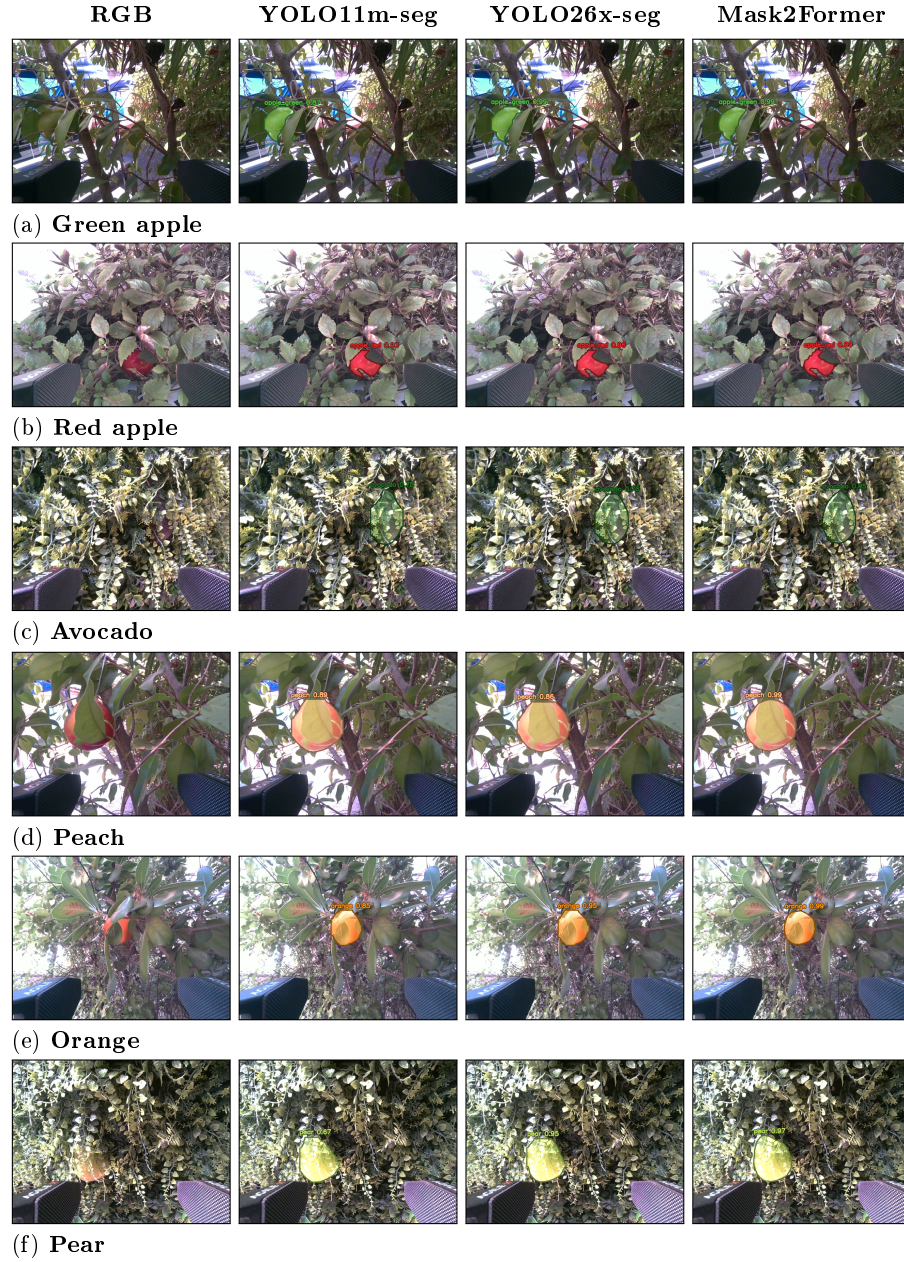


Fig. 4: Qualitative comparison of YOLO11m-seg, YOLO26x-seg, and Mask2Former under different fruit categories and occlusion conditions using identical RGB inputs. Mask2Former generally produces more coherent masks and improved instance separation in highly occluded regions, whereas YOLO11m-seg provides a favorable balance between segmentation quality and computational efficiency.

CNN-based approaches prioritize efficient localized feature extraction and multi-scale fusion, enabling fast inference with relatively low computational overhead. In contrast, transformer-based architectures improve global contextual reasoning through attention mechanisms capable of modeling long-range spatial dependencies, which becomes particularly advantageous under severe occlusion and cluttered scene conditions.

The visibility-aware evaluation suggests that transformer-based contextual reasoning improves the separation between partially visible fruits and surrounding vegetation in complex orchard scenes. In particular, Mask2Former exhibited more stable degradation behavior under reduced visibility conditions, indicating improved resilience in heavily occluded scenarios where fragmented observations and overlapping instances are common.

These observations further support the improved contour consistency and occlusion resilience of transformer-based segmentation. These differences were especially noticeable in cluttered scenes involving foliage occlusion, irregular object shapes, and strong background interference. In contrast, YOLO-based models occasionally produced fragmented or less consistent boundaries under severe occlusion, despite maintaining competitive inference speed and significantly lower computational cost.

Despite the qualitative and quantitative advantages of transformer-based segmentation, these architectures also introduce substantially higher model complexity and memory requirements. Such computational demands may still represent a limitation for deployment on lightweight robotic platforms or edge devices operating under strict real-time constraints. In this context, YOLO11-based models remain highly attractive for embedded agricultural robotic systems requiring efficient onboard inference while maintaining competitive segmentation performance.

Overall, the presented results highlight a practical trade-off between segmentation quality and computational efficiency for agricultural robotic perception. Transformer-based models provide improved contour consistency and stronger resilience under occlusion, whereas lightweight CNN-based approaches offer more favorable characteristics for real-time deployment and resource-constrained robotic applications.

The latency figures reported in Table 4 correspond to isolated model inference on a single workstation GPU, not to a complete robotic perception-to-actuation pipeline, which additionally involves sensor acquisition, pre/post-processing, grasp planning, and actuation overhead. Likewise, the models were not profiled on embedded or edge hardware. The real-time and embedded-deployment suitability claimed for the YOLO-based models in this work should accordingly be read as being based on model-level inference efficiency rather than on an end-to-end or on-device validation, which we identify as necessary future work.

One limitation of this study is that the evaluation was performed using a single dataset collected in a controlled environment rather than a working orchard, and at a fixed image resolution, with training and validation subsets only and no independent held-out test set – model selection and the reported metrics

therefore both come from the same validation split, and reported performance should be interpreted with this in mind until confirmed on unseen data. The controlled acquisition setting, while enabling repeatable multi-species data collection and a systematic visibility breakdown, does not capture every source of variability present in a working orchard (e.g. uncontrolled weather, seasonal foliage density, or fruit attached to the plant rather than arranged for capture), and performance on images captured directly in an operating orchard remains to be validated. Additional experiments across different crop types, environmental conditions, sensor configurations, and embedded hardware platforms, together with an independent test set and in-field data, would further strengthen the generalization analysis of the evaluated architectures.

## 6 Conclusions

This paper presented a comparative evaluation of YOLO11-seg, YOLO26-seg, and Mask2Former for multi-fruit instance segmentation under challenging, orchard-like foliage occlusion.

Experimental results showed that Mask2Former achieved the highest segmentation accuracy and improved robustness under severe occlusion, whereas YOLO11m provided the best balance between segmentation performance and inference efficiency for real-time robotic applications.

The presented benchmark highlights the complementary strengths of CNN-based and transformer-based architectures for agricultural robotic perception.

Future work will focus on integrating these models into complete robotic harvesting systems and investigating hybrid CNN-transformer approaches for improved robustness in dynamic agricultural environments.

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